Audiophile – Project Proposal

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Project Description

Audiophile is a proposed web-based system designed to support beginning to intermediate musicians during independent practice sessions. The goal of Audiophile is to help users improve pitch accuracy by combining real-time audio analysis with personalized feedback that clearly explains pitch, intonation, and scale patterns. The system focuses on instrument practice and is intended to provide immediate guidance without requiring instructor supervision.

Intended Use of the System

Audiophile is intended for users who want to learn more about music theory concepts in a practical, self-guided, and easy-to-understand way. The system is designed mainly for independent practice scenarios where learners want to verify whether they are playing notes correctly and in tune.

The system will be used by:

- Beginner and intermediate instrument learners,
- Music students practicing outside formal lessons,
- Hobbyist musicians seeking quick feedback.

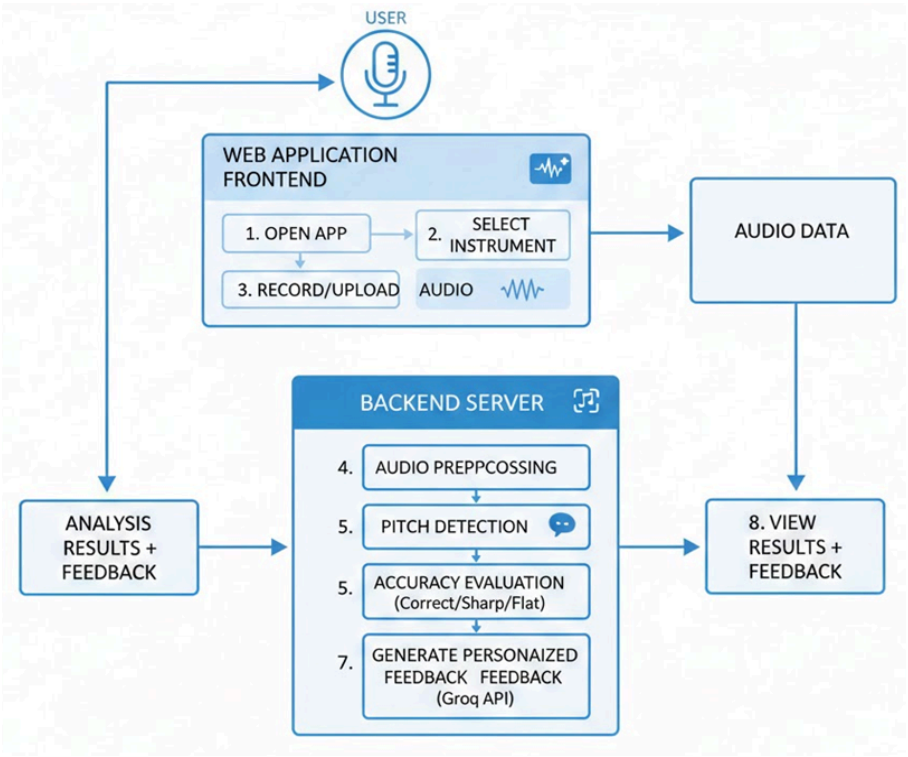
Overall Functionality of the System

Audiophile is designed to analyze instrument audio and provide meaningful feedback that helps users improve pitch accuracy and fundamentals in music theory. The system performs the following core functions:

- Accepts instrument audio through recording or file upload,
- Detects pitch from the audio signal,
- Identifies the musical note being played,
- Evaluates pitch accuracy as in tune, sharp, or flat,
- Matches the given pitch with the required pitch for scalar exercises,
- Generates personalized feedback using AI

Audiophile will support specific instruments to begin with to best serve those instruments and guide players through efficient ways of adjusting their pitch with the instrument, embouchure, or other techniques. By providing instant and easy-to-understand feedback, Audiophile helps users recognize pitch errors early and improve performance. AI-generated feedback avoids repetitive responses commonly seen in basic tuning applications and provides clearer guidance.

Architectural Breakdown:



Main Components of the System

Audiophile is designed using a modular architecture, where each component has a clear responsibility.

1. **User Interface Component:** Handles instrument selection, audio recording, file upload, and display of results. Separating the interface from processing logic ensures a clean user experience and easier frontend updates.
2. **Audio Input and Preprocessing Component:** Receives audio input and prepares it for analysis through normalization and formatting. Preprocessing improves pitch detection accuracy across different recordings.
3. **Pitch Detection and Evaluation Component:** Extracts pitch from audio, maps it to musical notes, and evaluates accuracy as in tune, sharp, or flat. Keeping pitch analysis separate allows focused development and easier testing.
4. **Instrument Validation Component:** Validates detected pitch using instrument-specific pitch ranges. This improves reliability since different instruments operate within different pitch limits.
5. **AI Feedback Generation Component:** Uses pitch accuracy results to generate personalized feedback through the Groq API. This component ensures feedback is varied, informative, and non-repetitive.
6. **Backend API Component:** Manages communication between frontend and backend components. API based communication ensures modularity and supports future system extensions.